Bagoo E-Commerce Platform

Multi-Tenant Logistics Architecture, Operational Flow & Earnings Model

Document Version: 2.0.0 | Architecture Design Document (ADD) | Conceptual Specification

1 System Overview

The Bagoo logistics engine connects buyers, sellers, couriers, and warehouse sorting centers. The system uses a **Mother Hub and Child Hub** model.

Parcels never move randomly between neighboring towns. Instead, parcels are picked up locally, brought to a nearby Child Hub, consolidated and sent to a central Mother Hub for regional sorting, transferred to the buyer's local Child Hub, and finally dispatched to a delivery rider.

2 Platform Roles and Responsibilities

The system defines five primary operational roles and one platform owner role:

- **Super Admin (Bagoo Platform):** Reviews and approves registered logistics companies, oversees dispute resolution, manages platform fee settings, and monitors the global financial ledger.
- **Logistics Company Admin:** The business owner of the logistics company. Once approved by the Super Admin, this admin sets up their physical network (Mother Hub and Child Hubs) and generates accounts for warehouse operators and riders.
- **Hub Staff / Operator:** Warehouse station workers who operate barcode scanners at physical stations to receive incoming parcels, sort packages into destination bins, pack items into Mother Bags, and dispatch trucks.
- **Rider / Courier:** Drivers who operate the mobile courier application. They perform first-mile store pickups from sellers and last-mile doorstep deliveries to buyers, collect Cash-on-Delivery (COD) payments, and take photo Proof of Delivery (POD).
- **Seller:** Packs purchased items, generates and attaches the Air Waybill (AWB) with the tracking barcode, and hands over parcels to the pickup rider.
- **Buyer:** Purchases products on the platform, chooses their preferred registered logistics courier during checkout, and tracks package status updates in real time.

3 End-to-End Product Flow: From Checkout to Doorstep

Below is the complete physical and data journey of a parcel using the Laguna-to-Batangas scenario (Seller in Biñan, Buyer in Pagsanjan):

1. **Order Placement and Routing Setup:** The buyer purchases a product and chooses Fast-Logistics. The system checks the seller's location (Biñan), the buyer's location (Pagsanjan), and maps both to their respective local Child Hubs. The system assigns a unique tracking number and pre-computes the route: Biñan Child Hub $\rightarrow$ Calamba Mother Hub $\rightarrow$ Santa Cruz Child Hub. The seller prints the Air Waybill (AWB) and attaches it to the parcel. *System Status:* READY_FOR_PICKUP.

2. **First-Mile Pickup Handover:** A pickup rider assigned to Biñan arrives at the seller's store. The rider opens the mobile courier app, switches to Pickup Mode, and scans the parcel barcode with the phone camera. The rider takes physical custody of the item. *System Status:* PICKED_UP.

3. **Origin Child Hub Receiving (Biñan Hub):** The pickup rider arrives at the local Biñan Child Hub and unloads all collected packages at the receiving dock. A station worker scans the barcode using a desktop scanner. This scan legally transfers custodial liability from the courier to the station. *System Status:* ARRIVED_AT_ORIGIN_CHILD_HUB.

4. **Container Bagging (The Mother Bag):** Hub workers group parcels heading toward the central region into a heavy-duty container sack called a Mother Bag. The worker scans the Mother Bag barcode, scans the parcel inside it, and seals the sack with a numbered tag. A feeder van driver scans the Mother Bag tag onto the van manifest and drives to the central facility. *System Status:* IN_TRANSIT_TO_MOTHER_HUB.

5. **Central Sorting at the Mother Hub (Calamba Hub):** The feeder van unloads at the Mother Hub. Staff cut the bag seal and scan each parcel individually. Upon scanning, the workstation monitor immediately flashes the assigned destination zone: **"EAST LAGUNA – SANTA CRUZ CHILD HUB"**. The worker drops the parcel into the Santa Cruz rolling cage. *System Status:* ARRIVED_AT_MOTHER_HUB.

6. **Line-Haul Trunk Transfer:** Parcels in the Santa Cruz cage are packed into an outbound line-haul sack. A heavy freight truck arrives at the Mother Hub dock, scans the master sack tag onto the trip manifest, and drives along the highway trunk line to East Laguna. *System Status:* IN_TRANSIT_TO_DESTINATION_CHILD_HUB.

7. **Destination Child Hub Inbound (Santa Cruz Hub):** The truck arrives at the Santa Cruz Child Hub early in the morning. Workers cut the seal and scan the parcel barcode. The screen directs the worker to place the package onto **"Shelf C – Pagsanjan Delivery Route"**. *System Status:* ARRIVED_AT_DESTINATION_CHILD_HUB.

8. **Last-Mile Dispatch:** The neighborhood delivery rider begins their shift. The rider opens the mobile app, selects Load for Delivery, and scans all packages located on Shelf C. Custody of the parcel and the responsibility for collecting payment are now transferred to this rider. *System Status:* OUT_FOR_DELIVERY.

9. **Doorstep Delivery and Escrow Release:** The rider arrives at the buyer's home in Pagsanjan. The rider collects the Cash-on-Delivery payment (if applicable), takes a clear photo Proof of Delivery (POD) showing the delivered parcel, and submits the delivery in the app. The system immediately marks the order complete and triggers automated earnings release. *System Status:* DELIVERED.

4 The Seven Core Logistics Algorithms

Logistics operations rely on seven core algorithmic concepts. These are explained conceptually below:

4.1 1. Forward Hierarchical Tree Routing

What it does: Determines the path of stations a package must travel through. When an order is placed, the system checks the seller's ZIP code to find the origin Child Hub, checks the buyer's ZIP code to find the destination Child Hub, and inserts the parent Mother Hub in the middle. The resulting hop sequence is stored directly on the parcel record.

4.2 2. Return-to-Sender (RTS) Reverse Engine

What it does: Handles delivery failures and cancellations. If a buyer rejects a COD parcel or is unreachable after three delivery attempts, the system cancels the forward delivery. It inverts the station path so the parcel flows in reverse: Destination Child Hub $\rightarrow$ Mother Hub $\rightarrow$ Origin Child Hub $\rightarrow$ Seller.

4.3 3. Misroute Trap and Recovery Engine

What it does: Prevents parcels from getting lost when a warehouse worker makes a physical mistake. Every time a barcode is scanned, the system checks: *"Does this current hub match the next scheduled stop in the parcel's route?"* If someone accidentally puts a Batangas package on a Cavite truck, the scanner sounds an error alarm and tells the worker to set the package aside for return to the Mother Hub.

4.4 4. Intra-Hub Bypass Optimization

What it does: Short-circuits the route for neighborhood deliveries. If the seller and the buyer live in the same town and share the same Child Hub, the algorithm detects that the origin hub and destination hub are identical. It removes the Mother Hub from the route, allowing the package to go straight from local intake to local delivery.

4.5 5. Volumetric Weight and Rate Engine

What it does: Prevents courier losses on light but bulky packages (like pillows or large plastic toys). The system measures parcel dimensions (Length $\times$ Width $\times$ Height in cm divided by 3,500) to find the volumetric weight. It compares this against the actual scale weight and calculates the shipping fee using whichever number is higher.

4.6 6. Containerization (Mother Bag Aggregation)

What it does: Speeds up warehouse operations by grouping dozens of small packages under one master barcode tag. When moving packages between hubs, workers scan only the master bag barcode rather than scanning 100 individual parcels. The database automatically cascades the transit update to every parcel inside that bag.

4.7 7. Rider Daily COD Cash Reconciliation Ledger

What it does: Ensures 100% cash accountability for couriers. The system calculates the exact cash sum of all COD packages marked "Delivered" by a rider during their shift. At the end of the day, the rider must hand over this exact physical cash amount to the hub manager before their shift can be closed.

5 Account Provisioning Workflow

To avoid confusion regarding company setup and access control, accounts are created in a strict order:

1. **Registration:** The business owner submits company registration details and valid permits to the Bagoo platform.
2. **Approval:** The Bagoo Super Admin reviews the legal documents and approves the company. The owner's account becomes a **Logistics Company Admin**.

3. **Hub Creation:** The Logistics Company Admin logs into their dashboard and creates their physical facilities (e.g., Mother Hub Calamba, Child Hub Biñan, Child Hub Santa Cruz).
4. **Staff Account Setup:** The Logistics Company Admin creates user accounts with login emails and temporary passwords for hub operators and delivery riders, linking each person to their assigned physical hub.

6 Financial Settlement and Earnings Breakdown

The system strictly separates **Product Subtotal** from **Shipping Fee**.

6.1 Baseline Scenario Metrics

- **Product Subtotal:** PHP 1,000.00
- **Fixed Flat Shipping Fee:** PHP 60.00
- **Total Amount Paid by Buyer: PHP 1,060.00**

6.2 Earnings Settlement Table (Standard Single-Rider Model)

Role	Allocation Rule	Amount (PHP)	Source Pool	Release
Buyer	Total Amount Paid	- 1,060.00	Customer Payment	Checkout
Seller	90% of Product Subtotal	+ 900.00	Product Subtotal	Buyer De
Bagoo Platform	10% Marketplace Commission	+ 100.00	Product Subtotal	Buyer De
Logistics Company	50% of Shipping Fee	+ 30.00	Shipping Fee	Successfu
Delivery Rider	50% of Shipping Fee	+ 30.00	Shipping Fee	Cash Rem
Super Admin	Platform Governance	0.00 direct	Bagoo Revenue Pool	Platform

Table 1: Earnings Distribution for PHP 1,000 Product with PHP 60 Shipping

6.3 Alternative Fleet Model: Dedicated Pickup + Last-Mile Split

In operations where the first-mile pickup driver and the last-mile delivery driver are separate roles, the PHP 60.00 shipping fee is partitioned as follows:

- **Logistics Company Hub Pool (PHP 25.00):** Retained by the company to cover warehouse sorting staff, hub building lease, and fuel for the line-haul inter-city trunk van.
- **Delivery Rider / Last-Mile (PHP 25.00):** Credited to the neighborhood courier upon successful doorstep delivery, cash collection, and valid photo Proof of Delivery.
- **Pickup Rider / First-Mile (PHP 10.00):** Credited to the collection courier once the parcel is scanned and received at the origin Child Hub.